Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Cincinnati, OH

SUMMARY

M.S. Computer Science candidate (Dec 2026) with production engineering experience building ML training/eval pipelines, LLM systems, and full-stack applications using Python, PyTorch, Java, and React. Current research builds a two-stage LLM pipeline over multimodal video and device-sensor data, with hands-on evaluation, prototyping, and demo-building across the ML lifecycle.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Machine Learning, Data Mining, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, C/C++, Swift, SQL, Go.

AI & ML: PyTorch, TensorFlow, Gemini, Vertex AI, scikit-learn, LangChain, LLM/Multimodal Foundation Models, Model Quantization, On-Device Optimization.

Web & Backend: React, Node.js, FastAPI, Flask, REST APIs.

Databases: PostgreSQL, MySQL, Redis.

Developer Tools: Git, Docker, Kubernetes, AWS, Google Cloud Platform, Linux, Bash, CI/CD.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Built end-to-end **ML training and evaluation pipeline** in Python/PyTorch-style tooling: designed two-stage memory-augmented LLM method decoupling multimodal video understanding from real-time device sensing/automation for Android GUI bug reproduction.
- Evaluated against **ViBR** (published GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ real-world apps using a provider-agnostic LLM abstraction (Gemini, GPT-4o, Llama, Qwen), isolating method effects from model choice.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity; built device automation harness with stall detection for real-time demo evaluation runs.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Built production software integrating infrastructure automation into Teamcenter X with Python (Boto3), reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing and resolving live infrastructure issues end-to-end.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, and GitLab CI/CD to modernize production software delivery.
- Served as Quality Coach, increasing automated code coverage from **35% to 78%**; mentored **5 engineers** through structured onboarding, strengthening cross-functional collaboration.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, PyTorch-style ML, scikit-learn, FastAPI] Jul 2026

- Built ML training/eval pipeline forecasting demand across **263 zones** (**2.3M-row** dataset); designed two-stage model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **multimodal RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling, built as production-quality inference infrastructure consumable by external AI agents.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).